6. Check control panel gauges for proper readings.
7. Adjust seat position and mirrors.
8. Allow the engine to warm and fasten your seat belt.
9. Disengage the park brake.

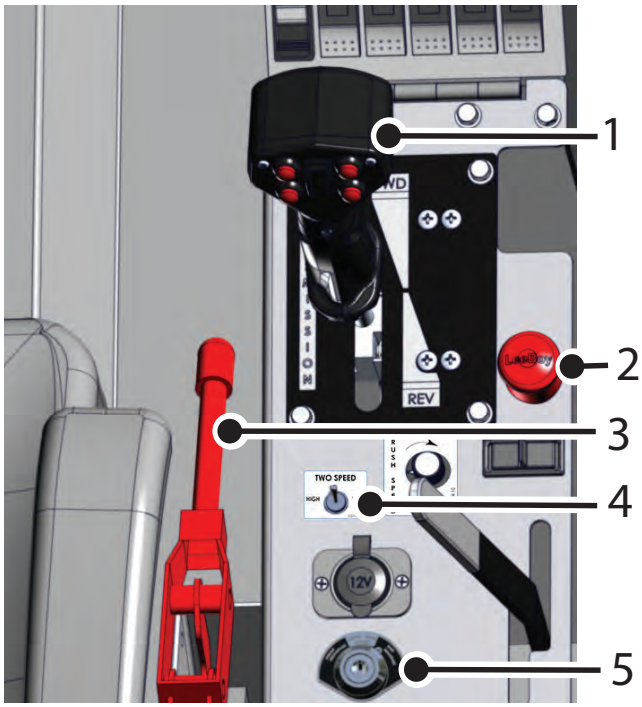


Figure 4-2. Start-Up Controls

1 - Joystick (Transmission Selector)

2 - Emergency Stop Button

3 - Park Brake Lever

4 - Speed Selector Switch

5 - Ignition Switch

Cold Weather Starting

- Engage the parking brake. **(Figure 4-2)**
- Place the transmission selector into the NEUTRAL position.
- Turn the ignition key to the Start position. The "Wait to Start" light will illuminate for a few seconds on the PV380 digital display controller.
- When the Wait to Start banner disappears, turn key to the ON position to crank engine.
- Allow the engine to idle about ten minutes before driving the machine.

Stopping the Engine

1. Adjust throttle to idle using the decrease RPMs feature on the PV380 digital display ***(Page 4-9)*** until idle speed is reached (1200 RPMs).
2. Apply park brake.
3. Turn ignition key counterclockwise to the OFF position and remove key.

NOTE: The electrical system will continue to operate for approximately 20 seconds after the ignition is turned off. This allows the engine ECU (Engine Control Unit) to store critical engine data as it powers down.

Emergency Stop Button

The purpose of the Emergency Stop Button (E-Stop) is to shut down the engine and all machine functions in an emergency situation. This additional safety feature should only be used if the operator needs to shut down all machine operations quickly. **(Figure 4-2)**



Figure 4-3. Driving the RB50

DRIVING THE BROOM

It is important that the operator become familiar with driving the Rosco RB50 Broom before operating it on the job site. Know and understand the control functions before operation. ***(Pages 3-6 and 3-7)***

While the RB50 Broom is easy to operate, operators should become accustomed to using the engine throttle, brakes and hydrostatic transmission before engaging the brush sweeper.

Keep these safety tips in mind when operating the broom:

- Perform an operational safety check before operating the broom, ensuring all equipment functions properly and all safety equipment (i.e., fire extinguisher) is present.
- **ALWAYS wear your seat belt.**
- **ALWAYS use turn signals.**
- While the RB50 Broom is easy to operate, operators should become accustomed to using the engine throttle, brakes and hydrostatic transmission before engaging the brush sweeper. The foot brake pedal is very aggressive!
- Stop at all railroad crossings and look both ways before proceeding.
- Watch your speed, especially if driving in bad road conditions. Use extra caution at intersections.
- Be aware of traffic behind you. Pull over to the side of the road to allow vehicles to safely pass if needed.
- Ensure proper warning flags and/or devices are in place to warn traffic of road construction, slow-moving vehicle or other traffic alerts. If working at night, ensure the work site is well-lit and well-marked.
- Use spotters to look out for danger when loading and unloading equipment from vehicles. Understand all communication signals used between equipment operators and workers on foot.
- Avoid obstacles that can damage the broom or cause an accident, i.e., manholes, signs, and road debris, etc.
- Vehicles parked on inclines should use appropriately-sized blocks placed behind or in front of the tires.
- Always be mindful of what's going on around you. Never allow anyone to walk behind or in front of the broom while operating, especially near the swing radius of the brush.

Cab Door Release

When you open the cab door completely on the RB50 Broom, the door will lock into place for operator convenience. Simply pull the door release knob to release the door from its locked position. **(Figure 4-4)**

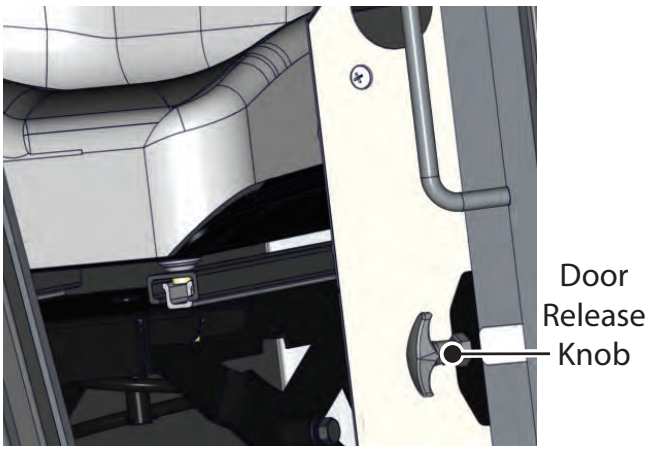


Figure 4-4. Cab Door Release Knob

Adjust Seat

The operator seat suspension provides comfort by absorbing some of the machine movements during operation. It can also be adjusted for individual comfort. **(Figure 4-5)**

Always make seat and seat belt adjustments BEFORE driving the sweeper. The seat belt is designed for safe operator retention if the machine is involved in an accident or roller-over.

- Use the suspension adjustment knob to set the suspension for operator weight. Rotate knob to increase or decrease setting as shown on the knob face.
- Use the seat position adjustment lever to move the seat forward or backward. Pull up on the lever to adjust and release to lock into place.



Figure 4-5. Seat Adjustment

Adjust Steering Console



Figure 4-6. Tilt Lock Toe Pedal

The steering wheel steers like any other steering wheel. Steering takes priority over brush operations (rotation, lift and swing) for operator safety.

You can adjust the steering column using the tilt lock pedal at the bottom of the steering column. **(Figure 4-6)**

To adjust, simply depress the pedal with your foot and tilt the steering column forward or backward to a comfortable position, then release the tilt lock pedal and it locks into place.

Cab AC/Heat

For added operator comfort, an air conditioning and heater option is also available. **(Figure 4-7)**

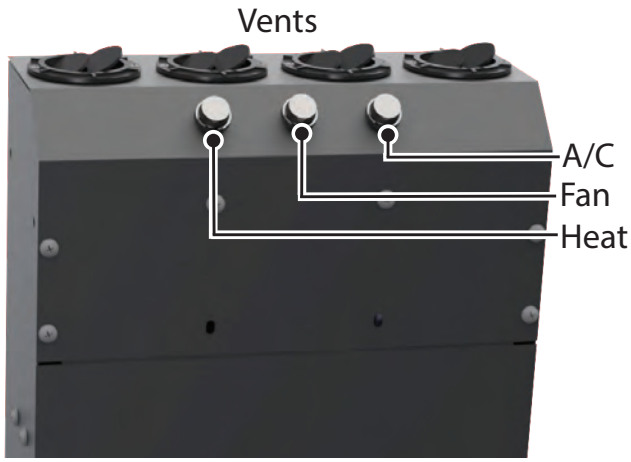


Figure 4-7. Cab AC/Heat

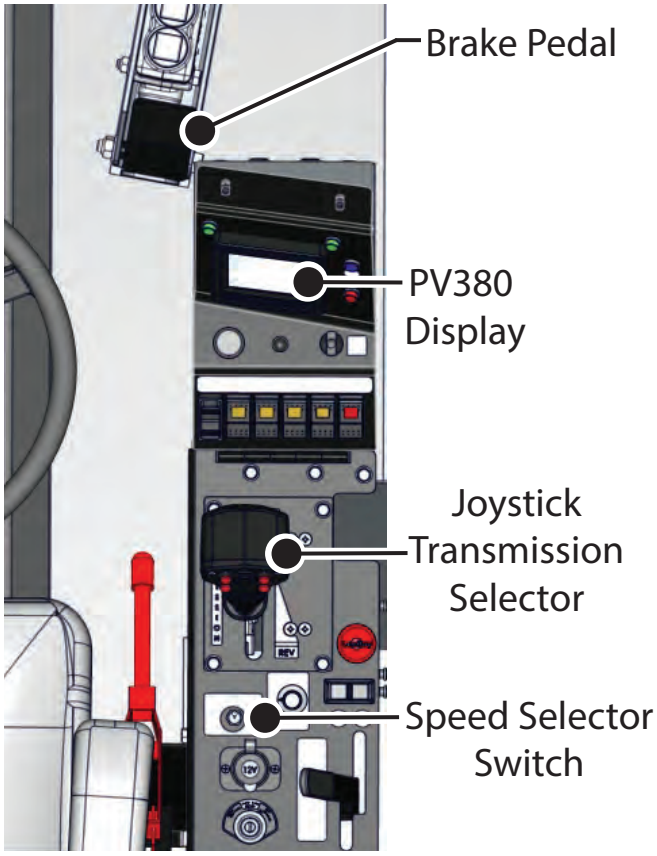


Figure 4-8. Operator Control Panel

Joystick Transmission Selector

The hydrostatic transmission selector joystick is on the control panel to the operator's right side. **(Figure 4-8)** With some experience, an operator can use this joystick lever along with proper throttle control to make smooth movements between forward and reverse. The center position is NEUTRAL.

- Push the joystick forward to move the broom forward.
- Pull the joystick backward to move the broom into reverse.

NOTE: Placing this joystick into reverse activates the back-up alarm to alert others of the machine's reverse motion.

Speed Selector Switch

The hydrostatic transmission is equipped with a HIGH/LOW Speed selector switch located on the right control panel. **(Figure 4-8)**

- Flip the switch left to the HIGH gear position for road travel.
- Flip the switch backward to the LOW position for sweeping operations.

⚠CAUTION NEVER operate in high gear while sweeping. The operator cannot drive and control the machine functions safely in high gear.

Stopping and Braking

The RB50 is equipped with a de-stroke braking system to provide aggressive stopping power in the event of an emergency. When the foot brake is applied, the hydrostatic pump is de-stroked, which redirects the flow and allows the foot brake to engage. **(Figure 4-8*)***

Though a brake pedal is situated on the floorboard to the right of the steering column, most operators seldom use it. Instead, operators use the throttle and hydrostatic transmission to slow down, stop or change direction. Become accustomed to using the hydrostatic transmission to assist in braking.

Regardless of the preferred stopping method, it is important to always put the transmission selector lever into the NEUTRAL position as the foot brake is applied or if throttling down to stop or turn.

⚠CAUTION The foot brake alone is not sufficient to stop the machine if stopping on a grade, especially steep grades. Familiarize yourself with the job site before sweeping to anticipate these variables.

Transmission Foot Pedal Option

Available for those who prefer is the optional forward and reverse foot pedal in lieu of joystick transmission control. **(Figure 4-9)** Adding the forward and reverse transmission foot pedal in lieu of the joystick control allows for a broader range of operator driving opportunities.

If equipped with this option:

- The **brake pedal** is on the left side and must be pressed to start the broom.
- The **transmission pedal** is on the right side and has a left and right pedal:
  - The left pedal moves the transmission into reverse.
  - The right pedal propels the broom forward.
  - Releasing the pedal puts the broom in neutral.

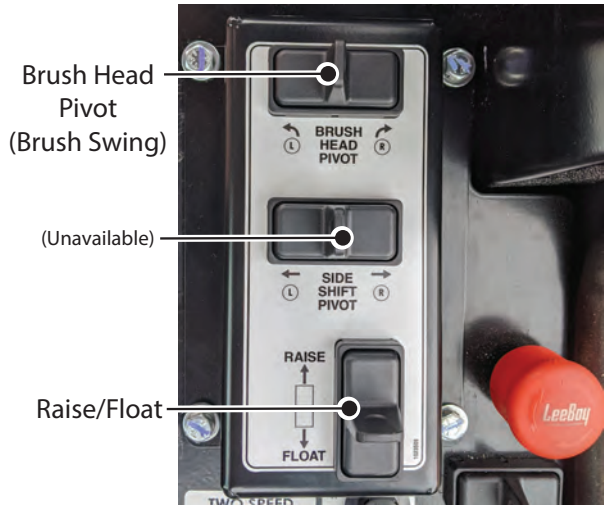


Figure 4-9. Control Panel Switches

Switches on the main control panel replace button controls on the joystick if equipped with this option.

- Push the brush head pivot switch to swing the brush left or right.
- Push the Raise/Float button forward or back for brush raise and float settings. (The side shift feature is not available for this model, therefore the switch is nonfunctional.)

PV380 POWERVIEW™ DISPLAY

The PV380 PowerView™ system is a multi-functional display that enables operators to view standard engine and transmission parameters along with active and stored trouble codes. The digital display screen can show single or multiple text descriptions, i.e., engine temperature, voltage, oil pressure, engine RPMs along with service reminders or fault conditions.

The operator can toggle through different screens for displaying critical data broadcast by the electronic engine and transmission Engine Control Unit (ECU).

Diagnostic codes with text descriptions of fault conditions are enhanced by ultra-bright alarm indicators that illuminate amber or red to alert the operator of a condition requiring attention. The amber warning light illuminates to visually alert the operator that a trouble code or service reminder is being shown on the display. If the red warning light illuminates, the operator should shut down machine and promptly repair the problem to prevent possible damage.

When turning the ignition key to the ON position, you will see a Murphy logo with the “Wait to Start Preheating” banner along the bottom of the screen. **(Figure 4-10)** DO NOT start the engine until this message disappears and the main gauge screen appears. **(Figure 4-11)**



Figure 4-10. Wait to Start Screen

Keypad Functions

The keypad is a touch system with five buttons along the bottom that corresponds to the navigational menu bar above the button (at the bottom of the display screen). **(Figure 4-11)** Button functions can change depending upon the screen displayed. When directed to press a symbol within the procedural steps, it is referring to the button below the displayed symbol on the navigational menu bar.



Figure 4-11. Navigation and Keypad Functions

Engine and transmission parameters displayed include:

- Engine Speed
- Engine Temperature
- Battery Voltage
- Coolant Temperature
- Engine Oil Pressure
- Total Machine Hours
- Engine Fuel Level
- Discharge Pressure
- Suction Pressure
- Active Fault Codes
- Stored Fault Codes

NOTE: Up to four (4) warning symbols can be shown at a time (upper left icon area). The most important symbol will be shown first followed by any others by priority.

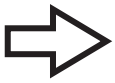
The following tables show functions that each button represents by the symbol appearing on the navigational menu bar and its function, depending upon the screen mode:

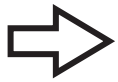
Table 4-4. Keypad Navigation Menu Bar Symbols

KEY 1	
Symbol	**Function**
⇦	Alternates between screen sets or moves the cursor one position to the left.
⇧	Scrolls up in certain screens when in specific menus.
-	Use to decrease numbers in various screens.

KEY 2	
Symbol	**Function**
FAULTS	Displays diagnostic messages (faults) and stored codes.
⇩	Scrolls down in certain screens when in specific menus.
+	Use to increase numbers in various screens.

KEY 3	
Symbol	**Function**
☰	Navigates to the Main Menu entrance point.
⇦	Serves as the Escape or Return to Previous Menu button.

KEY 4	
Symbol	**Function**
	Decreases the throttle set point.
	Decreases various settings.
STORED CODES	Displays the stored codes when in the diagnostic messages screen.
	Moves cursor one slot at a time to the right.

KEY 5	
Symbol	**Function**
	Increases the throttle set point.
ACTIVE FAULTS	Displays active faults.
	Serves as the “Enter” key for menu selections on certain screens.
	Increases various settings.
	Moves cursor one slot at a time to the right and displays alternate menu screens/choices.
	Moves the cursor to the next cell when customizing parameters on the home screen.
ACK	Acknowledges active faults.

Gauge Screens

Once the engine is running, the main gauge screen appears. There are four gauge screens--two electronic and two mechanical. You can scroll through the gauge screens by pushing ⇦ (Button 1). **(Figure 4-12)**

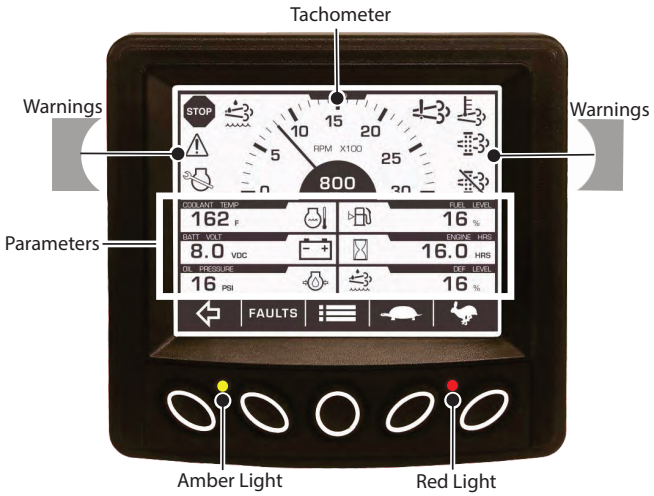


Figure 4-12. Gauge Screen Parameters

The gauge screen also shows the tachometer, engine information parameters and warning alerts.

Tachometer: The tachometer displays the engine speed on a 3000-RPM dial. Engine hours are displayed at the bottom of the tachometer area.

Parameters: The parameters sections of the screen displays engine oil pressure, engine temperature, engine hours, battery voltage, fuel level and other data. (This machine is not equipped with a DEF tank, therefore the DEF Level parameter does not function and always displays 0%.)

NOTE: The target speed gauge appears for five (5) seconds when the user throttles the engine.

Warnings: Alerts and warnings appear in this area of the screen to notify the operator a component needs attention. The most important warning symbols are shown in the upper left corner of the screen. Up to four (4) symbols can be shown at a time.

It is important to heed these warnings and resolve the issue promptly for optimum performance and preventing damage to the engine or other components.

When a warning message is displayed, an amber light will illuminate on the keypad. **(Figure 4-11)** If the red light illuminates, shut off the engine and resolve the problem to prevent possible damage.

User Settings Screen

These features allow you to customize , modify or make adjustments to screen items in your PV380 Digital Display. **(Figure 4-13)**

Navigation

To navigate to the User Settings screens:

1. Press the ☰ button to display the menu.
2. Use the ⇧ and ⇩ arrow buttons to scroll to the User Settings screen. **(Figure 4-13)**

Procedures follow for navigation and adjustment to each of the various User Setting functions.



Figure 4-13. User Settings Screen

Set Brightness and Contrast

You can select the desired screen brightness and contrast intensity using this option. (The display will automatically adjust the contrast when working at night.)

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Brightness and/or Contrast. (The selected function will be highlighted.)
3. Press + or - buttons to make adjustments within that selection.
4. Press ⏎ to save setting and exit the menu.

Set Language

The default language is set to English, but can be changed to German, Spanish, French or Italian.

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Language.
3. Press the ⇨ button to scroll to the desired language.
4. Press ⏎ to save setting and exit the menu.

Set Units

You may select how unit measurements are displayed using this option: US STD (U.S. Standard Measurements), METRIC KPA or METRIC BAR.

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Units.
3. Press the ⇨ button to scroll to the desired measurement units.
4. Press ⏎ to save setting and exit the menu.

Set Time

Follow these steps to set the time:

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Time. The cursor will be beneath the hour.
3. Use the + and - buttons to adjust the hour.
4. Press the ⇩ button to adjust the minutes (cursor will move right).
5. Repeat Steps 3 and 4 to adjust minutes (and seconds).
6. Press ⇦ to save setting and exit the menu.

Set Date

Follow these steps to set the date.

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Date. The cursor will be beneath the day.
3. Use the + and - buttons to adjust the day.
4. Press the ⇩ button and the cursor will move to the month.
5. Repeat Steps 3 and 4 to adjust the month and year.
6. Press ⇦ to save setting and exit the menu.

Show Clock

You can choose to show or hide the clock:

1. Navigate to the User Settings screen as described in the previous Navigation section.
2. Press the ↵ button to select Units.
3. Press the ⇨ button to choose Show or Hide.
4. Press ⇦ to save setting and exit the menu.

Service Reminders Screen

The Service Reminders screen allows the operator to scroll through the service list to view information about engine oil, hydraulic oil, air filter, fuel filter, service engine and service machine reminders. **(Figure 4-14)**



Figure 4-14. Service Reminders Screen

To navigate to the User Settings screens:

1. Press the ☰ button to display the menu.
2. Press the ⇩ button to scroll to Service Reminders.
3. Press the ↵ button to select Service Reminders.
4. Use the ⇨ and ⇦ buttons to scroll through the "read-only" screens.

Faults Screen

Faults in the system can be displayed by navigating to the Active Faults screen. **(Figure 4-15)**



Figure 4-15. Navigate to Faults Screen

Follow these steps to access the active faults screen.

1. Press the **FAULTS** button on Main Gauge screen. **(Figure 4-12 on *Page 4-11*)**
2. Use the ⇦ and ⇨ buttons to scroll through any messages present.
   - Each saved code shows the SPN (Suspect Parameter Number); FMI (Failure Mode Identifier); and OC (Occurrence Count), which indicates if the same fault occurred more than once. A text explanation of the warning or shutdown may also display.
   - To view the stored codes, press **Stored Codes.** "Requesting" and "Receiving Data" will be displayed while the system obtains the data. (If the data is not received, "TIMEOUT" will display and you will need to press the **Stored Codes** button again.)
3. Use the ⇧ and ⇩ buttons to scroll through any additional messages (if present). Press **Get Faults** to receive any additional stored faults.
4. Press ⇦| to exit and return to the main gauge screen.

PV380 Diagnostic Codes

**** PGN: Parameter Group Number**

Table 4-5. Diagnostic Trouble Codes (DTC) Chart

ICON (or Text)	PGN	DESCRIPTION
STOP	DM1	Stop Diagnostic Lamp: If this icon illuminates, there is an active stop fault. Stop the machine and resolve the problem before operating again to prevent damage.
⚠	DM1	Warning Diagnostic Lamp: If this icon illuminates, an active fault exists that needs immediate attention.
ACCEL PED1	61443	Accelerator Pedal Position 1
	61443	Percent Load at Current RPM
	61444	Actual Engine Torque
	61444	Engine Speed
TRIP DIST	65248	Trip Distance
VEH DIST	65428	Total Vehicle Distance
ENG HRS	65253	Total Engine Hours
TRIP FUEL	65257	Trip Fuel
FUEL USED	65257	Total Fuel Used
	65262	Engine Coolant Temperature
	65262	Fuel Temperature
	65262	Engine Oil Temperature
INTC TEMP	65262	Engine Intercooler Temperature
	65263	Fuel Delivery Pressure
	65263	Engine Oil Level
	65263	Engine Oil Pressure

ICON (or Text)	PGN	DESCRIPTION
	65263	Coolant Pressure
	65263	Coolant Level
	DM1	Service Indicator
VEH SPD	65265	Wheel-Based Vehicle Speed
FUEL RATE	65266	Fuel Rate
FUEL ECON	65266	Instantaneous Fuel Economy
AVG ECON	65266	Average Fuel Economy
BARO PRES	65269	Barometric Pressure
	65269	Air Inlet Temperature
BST PRES	65270	Boost Pressure
	65270	Intake Temperature
	65270	Air Filter Pressure
	65270	Exhaust Gas Temperature
ALT VOLT	65271	Alternator Voltage
	65271	Electrical Voltage
	65271	Battery Voltage
	65272	Transmission Oil Pressure
	65272	Transmission Oil Temperature
	65213	Fan Drive
AUX TEMP	65164	Auxiliary Temperature
AUX PRES	65164	Auxiliary Pressure
SLECT GEAR	61445	Selected Gear

ICON (or Text)	PGN	DESCRIPTION
CURNT GEAR	61445	Current Gear
OUT SFT SP	61442	Output Shaft Speed
IN SFT SPD	61442	Input Shaft Speed
TORQ LOCK	61442	Torque Converter Lockup
AUX IO 1	65241	Auxiliary IO Status 1
PEDAL SWT	61443	Accelerator Pedal Switch
DES ENG SP	65247	Desired Engine Speed
THROTTLE	65266	Throttle Position
	65270	Air Inlet Pressure
ENG TIMING	65159	Actual Engine Timing
ENG REVOLU	65253	Total Engine Revolutions
REQ GEAR	65256	Requested Gear
	65276	Fuel Level
	61448	Hydraulic Pressure
	65128	Hydraulic Temperature
	61444	Machine Hours
	Analog Input 1	Discharge Pressure
	Analog Input 2	Suction Pressure

4

STARTING TO SWEEP



Figure 4-16. Sweeping with the RB50 Broom

The Rosco RB50 Broom features excellent broom control and functionality for a variety of industrial and commercial applications. The broom operator controls are conveniently located on the control panel to the operator's right side. These are the primary controls used while sweeping. Refer to controls on ***Pages 3-6 and 3-7.***

This broom is easy to operate for drivers of all experience levels. Its heavy-gauge, rectangular tube frame and industrial-grade structure make the RB50 a road workhorse from general application sweeping to highway jobs. Add the optional front blade for mild pushing and/or optional curb brush to maximize productivity.

Joystick Controls

The transmission selector joystick is equipped with controls to lift or float the brush, and swing it left or right. **(Figure 4-17)** The operator can control these brush functions along with increasing or decreasing the machine's speed as needed for the job by pushing the joystick lever forward and in reverse. The brush speed is also controlled by changing the engine's RPMs using the joystick transmission control.



Figure 4-17. Joystick

Brush Lift/Float

The Brush Lift/Float Control switch is located in the center of the joystick. **(Figure 4-17)**

- Pushing the button up lifts the brush. Releasing the switch automatically returns the brush to the "hold" position.
- The center position "holds" or locks the brush carrier at the height last used while in the UP position.
- Pushing down on the switch activates the brush "float" mode. When releasing the switch back to center, float is maintained until the UP switch is pressed.

Brush Swing

Push the brush swing switches on the joystick slowly to gradually adjust the brush swing left or right. **(Figure 4-17)** Hold the button down to move the brush to its full swing position.

Brush Rotation Lever

The Brush Rotation lever is located on the control panel. **(Figure 4-18)**

- Pushing the lever forward allows the brush to push debris to the front of the broom. This is the normal sweeping position.
- Pulling the lever back turns the brush rotation off.

Brush Speed and Down Pressure

The Brush Down Pressure knob controls the amount of brush contact pressure to the work surface. **(Figure 4-18)**

- Turn dial left to increase brush speed.
- Turn dial right to decrease brush speed based on engine RPMs.
- Turn the dial clockwise to increase pressure.
- Turn the dial counterclockwise to decrease pressure.
- For best results, operate in the FLOAT position.

NOTICE **Be careful not to use excessive down pressure as the brush can be damaged.**

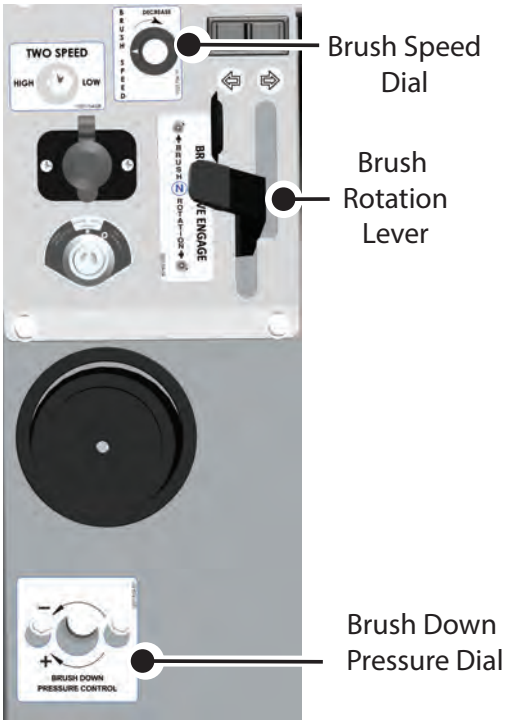


Figure 4-18. Brush Down Pressure Knob

SWEEPING GUIDELINES

An easy way to determine the ideal brush height setting is to sprinkle a little sand or dirt under the brush before beginning. Put transmission in neutral and set the park brake. Engage the brush using the Brush Rotation lever and press the Lift/Float control switch. Turn off the Forward/Reverse and raise the brush. There should be a clean swatch under the entire length of the brush.



Figure 4-19. Sweeping

- Sweep with the tips of the brush bristles for maximum results. Adjust the sweeping action using the Brush Down Pressure control valve to create a 3- to 4-inch wide footprint the width of the sweeper where the broom touches the pavement.

NOTICE **Excessive down pressure can wear and possibly damage the brush.**

- For most sweeping operations, the FLOAT position works best.
- For heavy sweeping, slow the ground speed and increase the brush RPMs. The right combination of ground speed and brush speed will correct ineffective sweeping.
- Engage the dust suppression system when sweeping in extremely dusty areas, especially when sweeping in high-traffic areas.

BRUSH CORE ADJUSTMENT

There are two adjustment screws on the brush frame for adjusting the brush core due to uneven brush wear or "coning." **(Figure 4-20)**

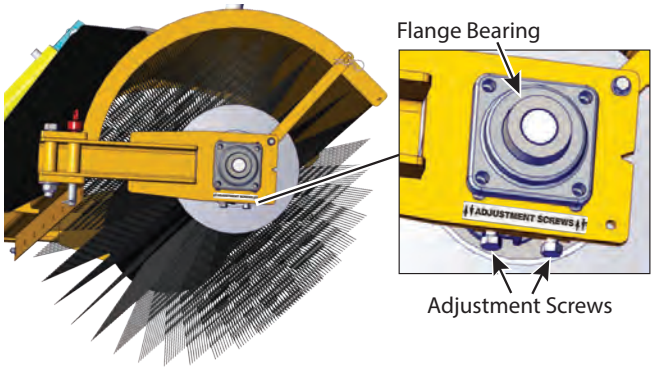


Figure 4-20. Brush Core Adjustment Screws

To make this adjustment:

1. Park vehicle on a flat surface and engage the parking brake.
2. Align the brush frame and brush core parallel to the front bumper.
3. Measure the height from the bottom of the brush bristles to the ground. **(Figure 4-21)**

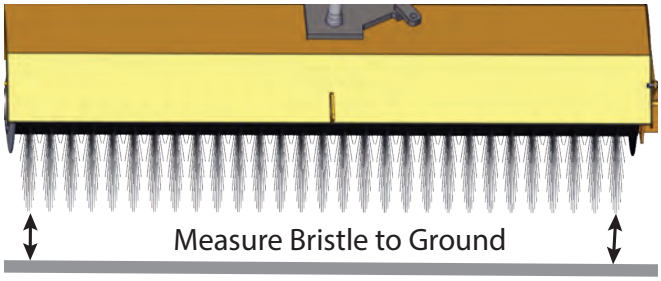


Figure 4-21. Measurement

4. Loosen the four (4) flange-bearing mounting bolts. DO NOT REMOVE.
5. Use the adjustment screws to bring the drive and swing arm sides of the core up or down, ensuring equal distance from the ground on both sides to level the brush core.
   - Turn bolt clockwise to raise height.
   - Turn bolt counterclockwise to lower height.
6. Once adjusted, retighten all four flange-bearing mounting bolts.

QUICK-CHANGE BRUSH CORE

The RB50 features a Quick-Change Core System that requires no tools to change the brush. Be sure to grease the brush core shaft bearings and lift pivot daily using a multi-purpose grease for optimum operation.

1. Park broom on flat surface, place machine in neutral, turn off the engine and set park brake.
2. Remove both cotter pins from brush cover. **(Figure 4-22)**

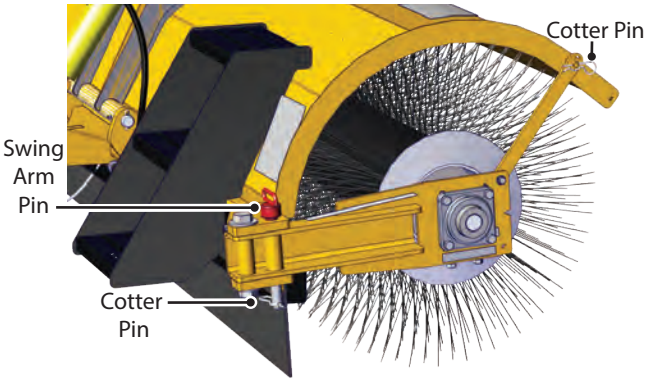


Figure 4-22. Brush Core Pins

3. Remove the swing arm pin.
4. Pull the brush core away from the drive hub and slide out. Set aside.
5. Check hub on both sides for damage. (Consult your authorized dealer if any damage is noted.)
6. Grease the U-Joint on the idle hub.
7. Slide new brush onto the slotted drive hub, ensuring the key stock inside the brush core goes onto the drive hub side.
8. Align the idle hub with the smooth end of the brush core and push the swing arm back into position into the brush core, keeping hands away from pinch points. Carefully guide the hub by hand if necessary to start into the core center.

NOTE: Brush carriage may need to be raised slightly to install a new core if the old brush core was worn.

9. Once the idle hub is fully engaged, reinsert the swing arm pin and cotter pins.

TRANSPORTING THE BROOM

Do not tow the RB50 to prevent damage to the machine. Transporting the broom on a flatbed truck (or trailer) is the preferred method of transporting the broom safely in all driving situations.

1. Clean the broom with a pressure washer, ensuring you remove all loose gravel, mud or debris from the wheels, frame and brush.
2. Ensure the combined height of the truck bed (or trailer) and loaded broom meeting height restrictions for the transport route, including bridges, overpasses and overhead obstructions.
3. Park the transport vehicle on a level surface and set the park brake.
4. Install chocks or blocks against the truck and trailer wheels.
5. Use a ramp or loading dock that has the capacity to support the broom, with a low angle of rise to the trailer bed.
6. Using the joystick controller, center the brush. Raise the brush to its full upright position.
7. Carefully and slowly drive the broom onto the trailer, centering the broom over the center line of the trailer.
8. Set the park brake and block brooms wheels in both directions with blocks or chocks to prevent rolling.
9. Lock brush in the UP position using the brush lock arm as shown below in **Figure 4-23.** To lock, move arm up as shown.



Figure 4-23. Brush Lock

10. Turn off the engine and remove key.
11. Fasten chains (or cables) to the broom frame using the four D-Ring tie-downs located on each side of the frame front and rear. **(Figure 4-24)**
    - Be careful you do not place chains or cables over the broom's hoses or electrical harnesses. Fasten chain or cable only to the D-Rings on the frame.

⚠WARNING To prevent the broom from breaking loose and being damaged in transport, ensure it is properly secured to the transport trailer.



Figure 4-24. D-Ring Tie Down

12. Cover the exhaust opening with heavy gauge plastic to prevent dust and moisture from entering the engine.
13. After transporting the machine, be sure to remove the plastic before operating.

OPTIONS

Brushes

The 7.6-feet brush options are Poly/Steel Tube, Poly Wafer, or Poly/Steel Water brushes. Contact your authorized Rosco LeeBoy dealer for purchase.

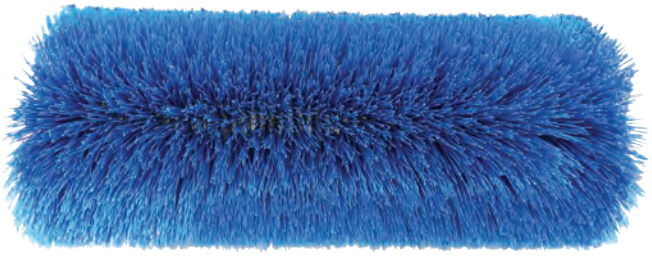


Figure 4-25. Brush Replacement Options

Front Windshield Washer and Rear Wiper/Washer

The front windshield wiper is standard, but the windshield washer is optional. A rear wiper with washer assembly is also an available option. See your authorized dealer for these options.

Rearview Camera System

A rearview camera system is also an available option. **(Figure 4-26)** This additional safety feature assists operators when moving in reverse by using the rear camera to see directly behind the machine.



Figure 4-26. Rearview Camera

Dust Suppression System

In certain conditions, it may be necessary to sweep using the Dust Suppression System to control dust. The Rosco RB50 is equipped with a 150-gallon water tank along with corrosion-resistant nozzles along the spray bar that spray water as the broom sweeps. **(Figure 4-27)** The spray bar runs the length of the broom. Be sure to inspect and clean the water nozzles after each use to ensure the spray pattern is not adversely affected.



Figure 4-27. Dust Suppression System

To operate the Dust Suppression system:

1. Fill up the water tank.
2. Turn on the Water System switch located on the control panel.
3. Follow the sweeping procedures and guidelines contained in this section.
4. Be sure to inspect and clean the water nozzles after each use to ensure the spray pattern is not adversely affected.

Cab Audio and Bluetooth

Equip your broom with AM/FM radio, CD player and Bluetooth audio system for operator enjoyment on the job.

Forward/Reverse Brush Rotation

Adding the ability for reverse brush rotation provides more flexibility for optimal cleaning.

Spare Tire and Mount

An optional spare tire with mount is also available. See your authorized dealer for purchase and installation.

Curb Brush

An optional curb brush attachment hydraulically adjusts for sweeping next to retaining walls, buildings, guardrails and curbs. It conveniently raises to a vertical position when not in use. **(Figure 4-28)** The curb brush features a 27-inch (68.6 cm) diameter heat-treated, steel brush with hydraulic motor drive. Refer to **Section 7,** *Illustrated Parts List (IPL),* for parts diagram.

1. Raise the brush to its full upright position and lock into place.
2. Turn off the engine and set the park brake.

NOTE: Be careful handling the curb brush as the bristles are sharp and could cause minor injury.



Figure 4-28. Curb Brush Attachment Option

Strike-Off Blade

You will need an assistant to attach and remove the optional strike-off blade attachment to the front of the broom. **(Figure 4-29)** Refer to **Section 7,** *Illustrated Parts List (IPL),* for parts diagram.

1. Park the broom on a level surface, turn off the engine and engage the parking brake.
2. Clean the blade-mounting front fender thoroughly before attaching blade.
3. Support the blade with jack stands in front of the broom front fender.

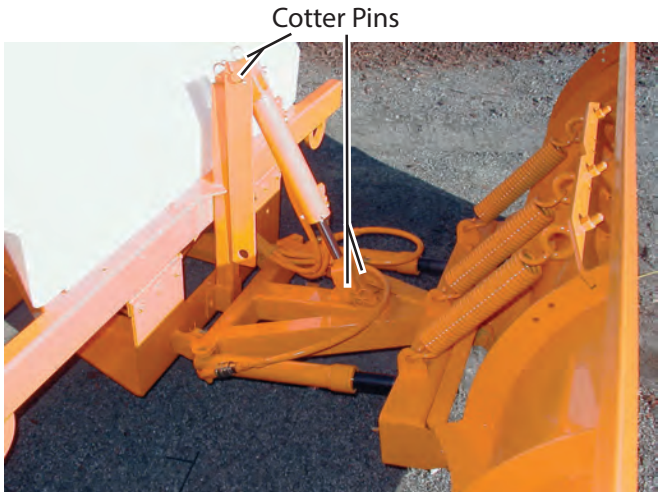


Figure 4-29. Blade Attachment Option



Figure 4-30. Telescoping Steering and Transmission Foot Pedal

Magnetic Cup Holder

Magnetic cup holders can be purchased for use with your road widener (and other machines in your fleet). **(Figure 4-31)**

See your authorized dealer for details. Ask for **Part Number 1020521.**



Figure 4-31. Magnetic Cup Holder

Telescoping Steering

The optional telescoping steering column coupled with the standard tilt column adds maximum cab comfort on the job. **(Figure 4-30)** The telescoping steering column option allows you to tilt the steering column and adjust your steering wheel for a more comfortable ride.

Cab Air Quality Kit

The Cab Air Quality Kit option offers the best line of defense for protecting your operators from exposure to harmful dust and toxins while sweeping. The RESPA®-CF2 delivers a powerful pre-cleaner, filter and pressurizer in a compact, custom unit that sustains positive pressurization and clean air in the cab.

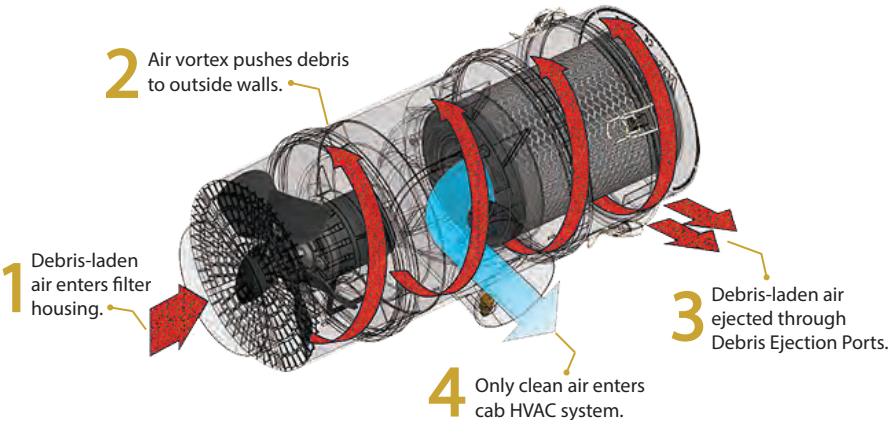


Figure 4-32. Vortex HyperFLOW® Technology

You will save maintenance costs with longer intervals for filter changes and less wear on the HVAC system. The self-cleaning air filter extends filter life up to 1,000 hours. Its air vortex HYPERFLOW® system continuously ejects particulate without collecting in filter housing.

Healthy, efficient and economical, the Cab Air Quality Kit can be easily installed on your broom. The system is recommended by the National Institute of Occupational Safety and Health. Contact your authorized dealer for purchase and retrofit.

Section 5

MAINTENANCE

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MAINTENANCE SCHEDULE

Before performing any maintenance procedures on the Rosco RB50 Broom, review the safety information in **Section 1**.

⚠WARNING Always use the appropriate and correct sized tools for the task at hand to prevent damage or possible injury.

⚠WARNING Always wear heat-resistant gloves when handling hot components.

Following the maintenance schedules and procedures will maintain the machine in top operating condition and provide years of trouble-free operation. Refer to the Engine Operator's Manual accompanying your machine for more detailed engine service information.

When performing any routine maintenance, always include the previous routine maintenance hours in the higher hourly schedule.

NOTICE Changing oil and cleaning the machine should only be done in a designated area where the oil and chemicals can be contained. These by-products should be discarded in accordance with environmental regulations.

Table 5-1. Periodic Maintenance Schedule

SYSTEM	ITEM	Every 10 Hours Daily	Every 50 Hours Weekly	Every 100 Hours Monthly	Every 250 Hours Quarterly	Every 500 Hours Semi-Annually	Every 1000 Hours Annually
Engine	Check oil level.	X					
	Check coolant level.	X					
	Replace coolant.					X	
	Check primary and secondary air filters.		X				
	Change air filters.			X			
	Replace fuel filter.				X		
	Check and drain fuel water separator.				X		
	Change oil and oil filter. **(Initial 50 Hours)**				X		
	Check main engine belt (Change every 3000 hours).			X			
Hydraulic	Check hydraulic oil level and filters.	X					
	Change hydraulic filters.				X		
	Change hydraulic oil.						X
	Clean hydraulic oil strainer.				X		
	Drain hydraulic tank. Replace suction strainer.						X
Mechanical	Lubricate steering axle pivot.			X			
	Lubricate steering axle king pins and tie rods.			X			
	Replace drive axle oil.						X
	Repack wheel bearings.						X
	Check fluid at master cylinder.	X					
	Change filter for crankcase bleeding.					X	

5

ROUTINE MAINTENANCE

Preparing Machine for Maintenance

1. Park the machine on a flat surface.
2. Engage the parking brake.
3. Turn off the broom drive.
4. Run engine at half-speed for three to five minutes, then reduce engine speed to a slow idle.
5. Lock brush in the UP position using the brush cylinder lock as shown below in **Figure 5-1.** To lock, move lock arm up as shown.



Figure 5-1. Cylinder Lock

6. Place ignition switch in the OFF position.

⚠WARNING If any maintenance must be performed while the engine is running, DO NOT leave the machine unattended.

Every 10 Hours or Daily Routine Maintenance

1. Lubricate the required parts. ***(Pages 5-6 to 5-8)***
2. Check battery cables and connections. ***(Page 5-13)***
3. Check coolant level. Fill if needed.
4. Check for indications of oil leakage. Check for loose, worn or damaged hoses, tubes and clamps. Tighten, repair or replace promptly.
5. Keep the fuel tank full to prevent condensation. Always fill at the end of the day.
6. Inspect the hand spray wand nozzle (and spray bar nozzles if equipped) to ensure they are free from clogs and operating properly.
7. Check the tires for proper inflation before operating the broom. ***(Pages 5-22 to 5-23)***
8. Grease the brush core shaft bearings and brush lift bushings. ***(Pages 5-6 to 5-8)***
9. Check engine oil level. ***(Page 5-9)***
10. Check hydraulic oil level and filters. ***(Page 5-18)***
11. Check hydraulic fluid at master cylinder in cab on the floor. ***(Page 5-25)***
12. Perform any other engine preventive maintenance described in the Engine Operator's Manual.

Every 50 Hours or Weekly Routine Maintenance

1. The FIRST 50 HOURS, change engine oil and oil filter, then every 250 hours thereafter. ***(Page 5-11)***
2. Lubricate the steering axle king pins and tie rods. ***(Pages 5-6 to 5-8)***
3. Check all battery connections and remove any corrosion present. ***(Page 5-13)***
4. Check air filter. Clean or replace if needed. ***(Page 5-12)***
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 100 Hour or Monthly Maintenance

1. Check main engine belt. Replace if needed.
2. Change air filter. ***(Page 5-11)***
3. Lubricate steering axle pivot. ***(Pages 5-6 to 5-8)***
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 250 Hours or Quarterly Routine Maintenance

1. Clean hydraulic oil strainer. ***(Page 5-19)***
2. Change the hydraulic oil and charge filter. ***(Page 5-18)***
3. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 500 Hours or Semi-Annual Routine Maintenance

1. Drain and replace coolant.
2. Change engine oil and filter. ***(Page 5-10)***
3. Replace fuel filter. ***(Page 5-12)***
4. Check and drain fuel water separator. ***(Page 5-13)*** Replace filter if needed.
5. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

Every 1000 Hours or Annual Routine Maintenance

1. Replace strainer in the hydraulic tank. ***(Page 5-19)***
2. Replace drive axle oil. ***(Page 5-25)***
3. Repack front wheel bearings. ***(Page 5-24)***
4. Perform previous routine maintenance and any other engine preventive maintenance described in the Engine Operator's Manual.

MACHINE LUBRICATION

Proper lubrication is necessary to maintain the machine at top efficiency. Use a high-quality multi-purpose grease for optimum performance.

(See the following pages for lubrication points.)

Be sure to grease the brush core shaft bearing and lift pivot daily.

It is important to grease the king pin bushings, tie rod ends, steering cylinder rod and steering axle pivot bushings every 100 hours of service.

⚠WARNING Use jack stands on a solid, flat surface to prevent the machine from tipping when it is raised. Block the remaining wheels using sturdy wheel chocks. DO NOT lie under the broom while supported only by a jack. Death or serious injury can result from improperly supporting a raised machine.

Use the following procedure for lubricating these components:

1. Locate and clean the single grease fitting at each of the lubrication points.
2. Apply one shot of grease to each fitting with a manual (hand) grease gun.
3. Wipe away any excess grease.

5

Lubrication Points

Lubrication points are shown below:

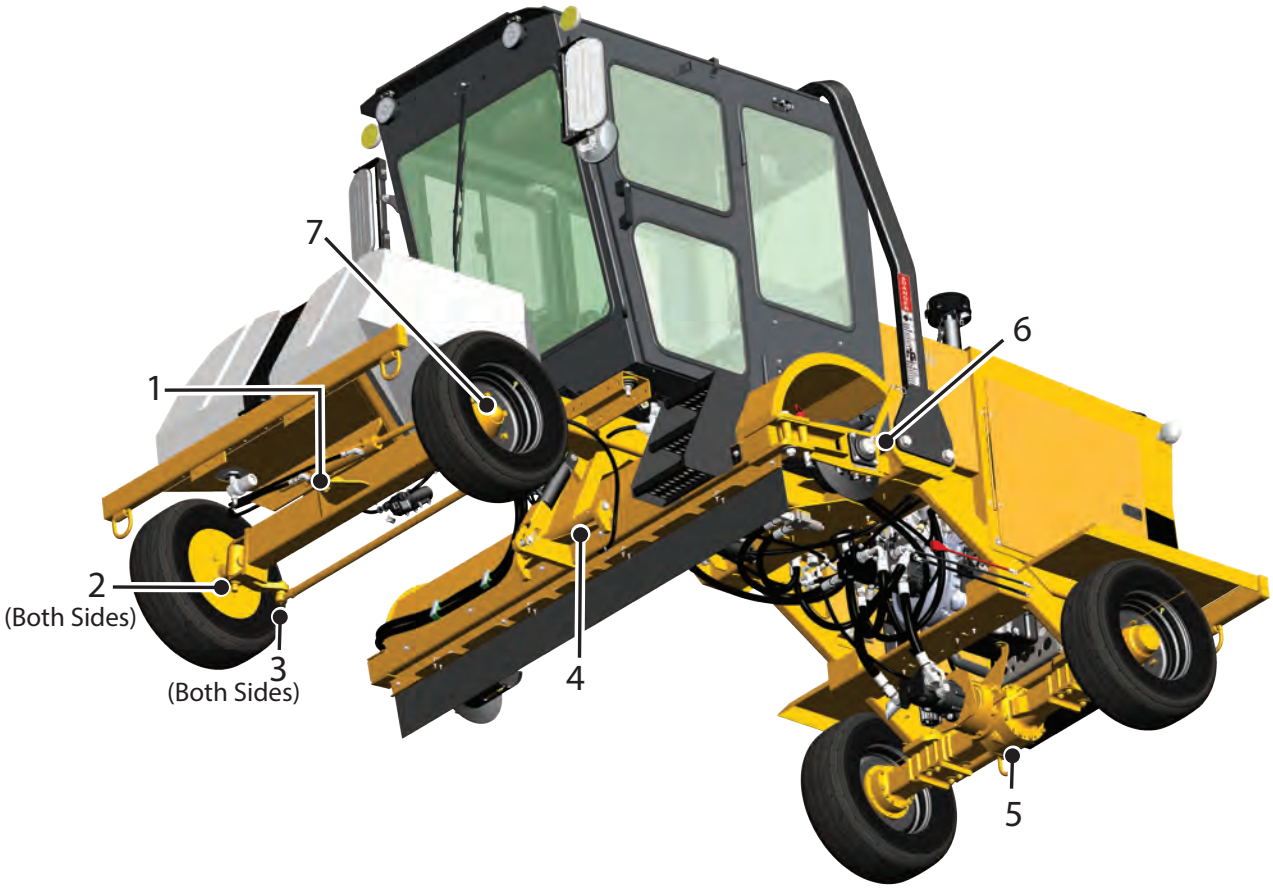
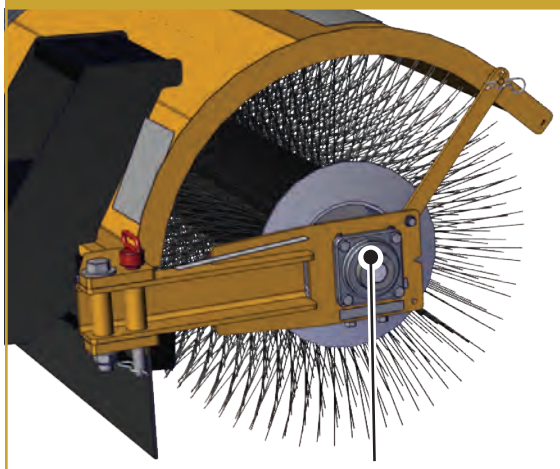
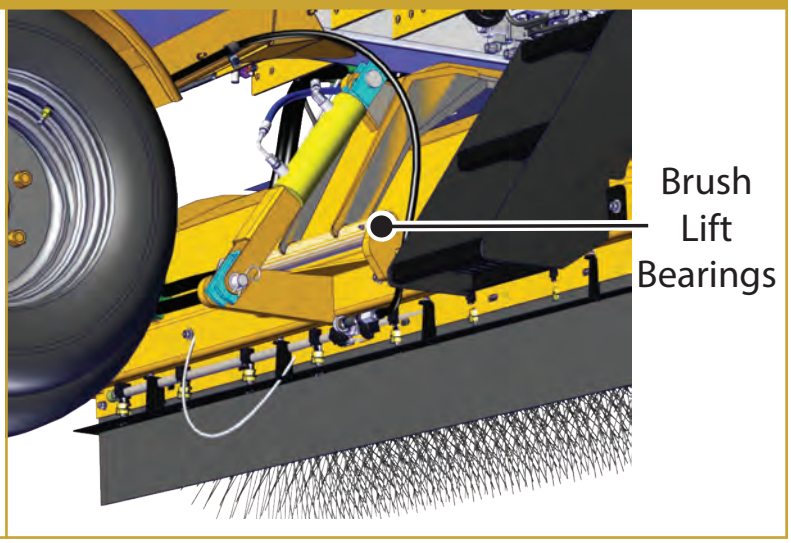


Figure 5-2. Lubrication Points Schedule

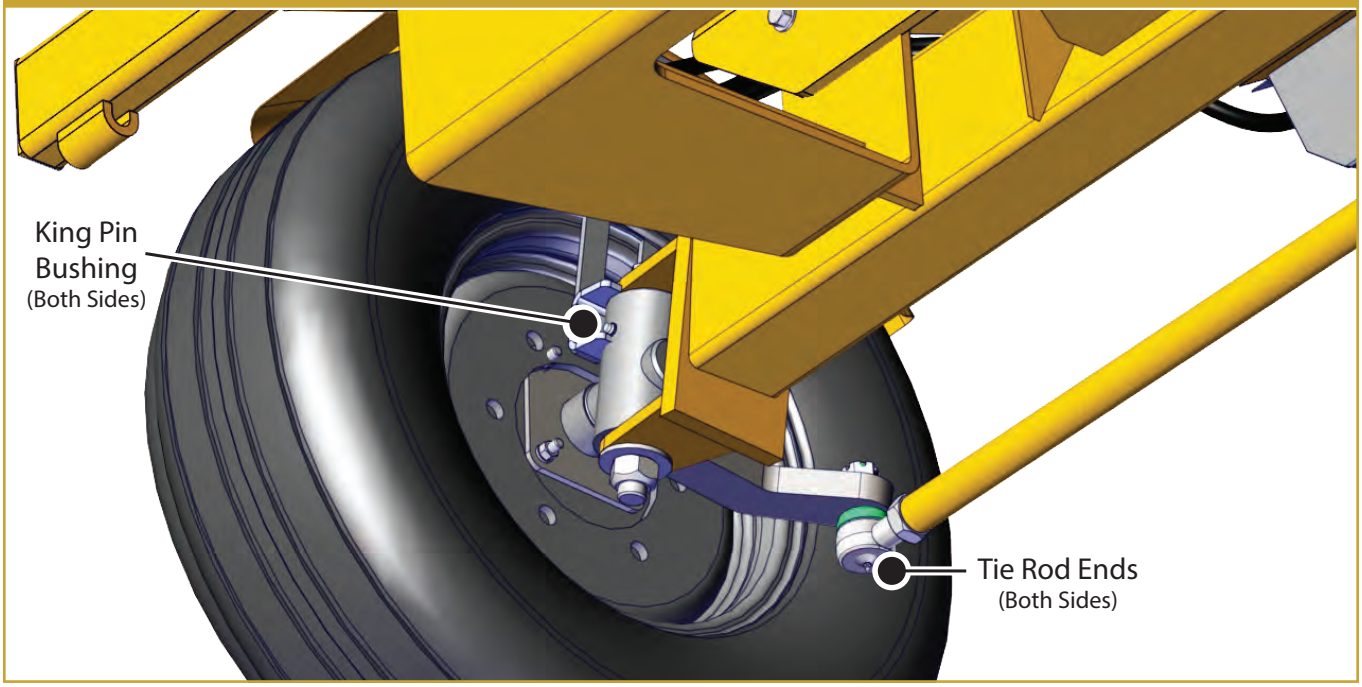
No.	ITEM	TYPE OF LUBRICANT	INTERVAL
1	Steering Axle Pivot	Multi-Purpose Grease	Monthly
2	King Pin Bushings	Multi-Purpose Grease	Weekly
3	Tie Rod Ends	Multi-Purpose Grease	Weekly
4	Brush Lift Bushings	Multi-Purpose Grease	Daily
5	Drive Axle	Gear Oil	Every 6 Months
6	Brush Core Shaft Bearings	Multi-Purpose Grease	Daily
7	Steering Axle Bearings	Multi-Purpose Grease	Every 6 Months

Daily Lubrication





Weekly Lubrication



5